

The Cubical King

From Worpitzky's Power Sums to a Three-Dimensional Delannoy Lattice

Counting Higher-Dimensional Lattice Walks via Worpitzky-Style Decomposition

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Abstract

Worpitzky's triangle encodes the coefficients that convert a sum of m -th powers into a sum of binomial coefficients, via the identity $\sum_{i=1}^n i^m = \sum_k W(m, k) \binom{n+1}{k+1}$. This makes it the natural combinatorial tool for counting structures organised by dimension: the squares on an $n \times n$ chessboard ($1^2 + 2^2 + \dots + n^2$), the small cubes in an $n \times n \times n$ block ($1^3 + 2^3 + \dots + n^3$), and so on. The Delannoy triangle counts the paths of a chess king from one corner of a board to the opposite, using three forward step types. The king moves through a cubical lattice and his admissible step types are organised by Worpitzky-style power-counting. We compute the central three-dimensional Delannoy numbers 1, 13, 409, 16081, $\dots$, derive their algebraic growth rate, and exhibit a closed-form decomposition by path length that mirrors Worpitzky's role in two dimensions. The growth-rate ladder reveals a clean structural pattern: the per-step expansion factor in d dimensions satisfies $(\alpha_d + 1)^d = 2\alpha_d^d$, with binomial coefficients appearing as the natural Worpitzky-style weights on the characteristic polynomial. The chessboard generalises crisply to the cubical lattice, and the algebra accommodates it.

1 The King's Promotion

Consider in two dimensions a king in traversing diagonally opposite corners of a chess board, driven inexorably forward via three legal step types: East, North, North-East. In three dimension chess he gains four more:

three single-axis steps: $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$

three double-axis steps: $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$

one triple-axis step: $(1, 1, 1)$

The single-axis steps have Euclidean length 1, the double-axis steps $\sqrt{2}$, the triple-axis step $\sqrt{3}$. There are $7 = 2^3 - 1$ step types in total: every non-empty subset of $\{x, y, z\}$ corresponds to one legal forward step.

This pattern is general. In d dimensions the king has $2^d - 1$ legal forward step types, organised by how many axes they advance. Worpitzky's organising principle (decomposing a

power-sum into a weighted binomial sum) is exactly the right book-keeping for this, and the structural reward is a clean algebraic ladder for the growth rates. We work the three-dimensional case in detail.

2 Matrices for Pascal, Worpitzky, and Delannoy

Before turning to the cubical king, we look at the three parent triangles in their familiar form and then see what each recurrence looks like when written as a matrix. The matrix view may be less familiar to readers, so we develop it from the ground up.

Pascal: the simplest case

Pascal's triangle, in its standard centred form, reads:

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 1 & & 4 & & 6 & & 4 & & 1 \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1
 \end{array}$$

Each entry is the sum of the two immediately above. Writing entries as $P(n, k)$ where n is the row number and k is the position within the row (counting from 0), the recurrence is

$$P(n, k) = P(n-1, k-1) + P(n-1, k).$$

Now consider how we generate row n from row $n-1$. Think of each row as a vector. Row 3 is the vector $(1, 3, 3, 1)$, row 4 is the vector $(1, 4, 6, 4, 1)$. The transition is

$$\begin{aligned}
 P(4, 0) &= P(3, 0) &&= 1 \\
 P(4, 1) &= P(3, 0) + P(3, 1) &&= 1 + 3 = 4 \\
 P(4, 2) &= P(3, 1) + P(3, 2) &&= 3 + 3 = 6 \\
 P(4, 3) &= P(3, 2) + P(3, 3) &&= 3 + 1 = 4 \\
 P(4, 4) &= P(3, 3) &&= 1
 \end{aligned}$$

Each new entry is a fixed linear combination of the previous row's entries. Linear combinations of a fixed vector are matrix multiplications. We can therefore write the passage from row 3 to row 4 as

$$\begin{pmatrix} 1 \\ 4 \\ 6 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 3 \\ 1 \end{pmatrix}.$$

The matrix is rectangular (5×4), because row n has one more entry than row $n-1$. The pattern of entries in this matrix is what mathematicians call *bidiagonal*: there are two diagonals of 1s, one shifted slightly below the other. Each column has exactly two non-zero entries, both equal to 1, encoding the rule "each new entry is the sum of its two parents." The number 1, 1 records both the *count* of parents (two) and the *weight* each gets (equal, namely 1).

Worpitzky: the same shape, with weights

The Worpitzky triangle, in its centred form, reads:

$$\begin{array}{cccccccc}
 & & & & 1 & & & \\
 & & & & 1 & & 1 & \\
 & & & 1 & & 3 & & 2 \\
 & & 1 & & 7 & & 12 & & 6 \\
 & 1 & & 15 & & 50 & & 60 & & 24 \\
 1 & & 31 & & 180 & & 390 & & 360 & & 120
 \end{array}$$

The recurrence is

$$W(n, k) = (k + 1) \cdot W(n - 1, k) + k \cdot W(n - 1, k - 1).$$

The shape is the same as Pascal's (two parents), but the weights are no longer equal to 1. The parent at position k contributes weight $k + 1$ and the parent at position $k - 1$ contributes weight k . As an example, from row 3 (1, 7, 12, 6) to row 4:

$$\begin{aligned}
 W(4, 0) &= 1 \cdot W(3, 0) &&= 1 \cdot 1 = 1 \\
 W(4, 1) &= 2 \cdot W(3, 1) + 1 \cdot W(3, 0) &&= 2 \cdot 7 + 1 \cdot 1 = 15 \\
 W(4, 2) &= 3 \cdot W(3, 2) + 2 \cdot W(3, 1) &&= 3 \cdot 12 + 2 \cdot 7 = 50 \\
 W(4, 3) &= 4 \cdot W(3, 3) + 3 \cdot W(3, 2) &&= 4 \cdot 6 + 3 \cdot 12 = 60 \\
 W(4, 4) &= 4 \cdot W(3, 3) &&= 4 \cdot 6 = 24
 \end{aligned}$$

producing row 4 = (1, 15, 50, 60, 24) as expected. In matrix form:

$$\begin{pmatrix} 1 \\ 15 \\ 50 \\ 60 \\ 24 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 7 \\ 12 \\ 6 \end{pmatrix}.$$

The pattern of entries is the same bidiagonal shape, but the values are no longer all 1. The lower entry in each column is k (the source column index), and the upper is $k + 1$. The asymmetric weighting is what makes Worpitzky's rows record the coefficients needed for the power-sum identity

$$1^m + 2^m + \cdots + n^m = \sum_{k=0}^m W(m, k) \binom{n+1}{k+1},$$

For example, on an $n \times n$ chessboard the total number of squares (of all sizes from 1×1 up to $n \times n$) is $1^2 + 2^2 + \cdots + n^2$, which for $n = 8$ gives 204. The analogous count of cubes of all sizes inside an $n \times n \times n$ block is $1^3 + 2^3 + \cdots + n^3$, giving 1296 for $n = 8$. Worpitzky's identity expresses both sums as weighted combinations of binomial coefficients, with the weights drawn from the rows of the Worpitzky triangle.

Delannoy: one more parent, equal weights

The two-dimensional Delannoy triangle, in its rectangular form, has entries $D(n, k)$ for $0 \leq n, k$ with the recurrence

$$D(n, k) = D(n - 1, k - 1) + D(n - 1, k) + D(n, k - 1).$$

Three parents instead of two. The first few values:

$n \backslash k$	0	1	2	3	4
0	1	1	1	1	1
1	1	3	5	7	9
2	1	5	13	25	41
3	1	7	25	63	129
4	1	9	41	129	321

The third parent ($D(n, k - 1)$, the same-row predecessor) complicates the matrix presentation a little, because it means row n depends on itself as well as on row $n - 1$. To write a clean row-to-row matrix we first eliminate the same-row dependency by substituting the recurrence into itself. The result is a relation of the form

$$D(n, k) = (\text{linear combination of row } n - 1).$$

For example, $D(2, 2) = D(2, 1) + D(1, 2) + D(1, 1)$. Substituting $D(2, 1) = D(2, 0) + D(1, 1) + D(1, 0)$ and $D(2, 0) = D(1, 0)$,

$$D(2, 2) = 2D(1, 0) + 2D(1, 1) + D(1, 2),$$

which gives $2 \cdot 1 + 2 \cdot 3 + 5 = 13$ as required. More generally row n is a linear combination of row $n - 1$ with *accumulating* coefficients, because the same-row dependency allows entries earlier in the same row to contribute multiple times to later entries. The coefficient pattern turns out to be remarkably simple:

$$M[k][j] = \begin{cases} 2 & \text{if } j < k \\ 1 & \text{if } j = k \\ 0 & \text{if } j > k. \end{cases}$$

The substitution matrix has 1s on the diagonal and 2s strictly below it:

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 2 & 1 & 0 & 0 & \cdots \\ 2 & 2 & 1 & 0 & \cdots \\ 2 & 2 & 2 & 1 & \cdots \\ \vdots & & & & \ddots \end{pmatrix}$$

The reader can verify by direct multiplication that M applied to row 1 = (1, 3, 5, 7, 9, ...) of the Delannoy table produces row 2 = (1, 5, 13, 25, 41, ...). The pattern of 1s on the diagonal and 2s below is the algebraic shadow of the three-parent recurrence: the diagonal 1 counts the direct contribution from $D(n - 1, k)$, while the 2s below count the combined contribution from the diagonal step $D(n - 1, k - 1)$ and the recursively-substituted same-row predecessor $D(n, k - 1)$.

What matters for our purposes is not this specific matrix but the *shape* of the Delannoy recurrence: three parents, equal weights 1, 1, 1. Pascal has two parents and weights 1, 1. Worpitzky has two parents and weights $k, k + 1$. The structural slots are *number of parents* and *values of the weights*, and the three triangles populate two of the four corners:

	Two parents	Three parents
Equal weights	Pascal	Delannoy (2D)
Weighted	Worpitzky	?

The bottom-right corner is where the cubical king naturally lives, but not as a two-dimensional triangle. The cubical king extends the “three parents” structure into a third spatial dimension, gaining four additional step types in the process. The weighting that emerges from this extension is not the position-dependent weighting of Worpitzky but a *dimension-dependent* weighting: each step type carries a binomial coefficient that counts how many ways it could be chosen from the available axes. The Worpitzky-flavoured promotion is therefore from a 2D weighted triangle to a 3D unweighted one, with the weighting absorbed into the geometry of the lattice itself.

This is what we will see in the next section, and it is the genuine three-dimensional extension of the chessboard king-walk problem.

3 The Cubical King

Place the king at the origin of an $n \times n \times n$ cubical lattice. He is to walk to the diagonally opposite corner (n, n, n) using only forward moves. The seven legal step types are listed above. Let $D_3(n_1, n_2, n_3)$ denote the number of distinct walks from $(0, 0, 0)$ to (n_1, n_2, n_3) . The recurrence is the natural three-dimensional Delannoy:

$$\begin{aligned} D_3(n_1, n_2, n_3) = & D_3(n_1 - 1, n_2, n_3) + D_3(n_1, n_2 - 1, n_3) + D_3(n_1, n_2, n_3 - 1) \\ & + D_3(n_1 - 1, n_2 - 1, n_3) + D_3(n_1 - 1, n_2, n_3 - 1) + D_3(n_1, n_2 - 1, n_3 - 1) \\ & + D_3(n_1 - 1, n_2 - 1, n_3 - 1) \end{aligned} \quad (1)$$

with $D_3(0, 0, 0) = 1$ and $D_3 = 0$ whenever any coordinate is negative.

The interesting numbers are the central values $D_3(n, n, n)$, the number of corner-to-corner walks:

$$D_3(n, n, n) = 1, 13, 409, 16081, 699121, 32193253, 1538743249, 75494983297, \dots \quad (2)$$

For comparison, the 2D central Delannoy, [4, 1] spine is 1, 3, 13, 63, 321, 1683, 8989, 48639, ..., smaller by a roughly exponential factor at each step.

4 Algebraic Growth Rate

To extract the growth rate, encode the recurrence in a trivariate generating function. With variables x, y, z tracking the three coordinates,

$$\sum_{n_1, n_2, n_3 \geq 0} D_3(n_1, n_2, n_3) x^{n_1} y^{n_2} z^{n_3} = \frac{1}{1 - x - y - z - xy - xz - yz - xyz}.$$

The denominator records the seven legal step monomials, one for each non-empty subset of the variables $\{x, y, z\}$.

The central diagonal is governed by the symmetric critical point obtained by setting $x = y = z = t$. This gives

$$\frac{1}{1 - 3t - 3t^2 - t^3}.$$

The coefficients of 3, 3, 1 in the denominator are the *binomial coefficients* $\binom{3}{1}, \binom{3}{2}, \binom{3}{3}$ that count the number of step types of each “dimensionality.” Three ways to single-step, three ways to double-step, one way to triple-step. The denominator factorises cleanly:

$$1 - 3t - 3t^2 - t^3 = 1 - ((1 + t)^3 - 1) = 2 - (1 + t)^3.$$

So the denominator vanishes when

$$(1+t)^3 = 2, \quad \text{hence} \quad t_* = \sqrt[3]{2} - 1 \approx 0.2599.$$

The radius of convergence is $\sqrt[3]{2} - 1$, the inverse is

$$\alpha_3 = \frac{1}{\sqrt[3]{2} - 1} = 1 + \sqrt[3]{2} + \sqrt[3]{4} \approx 3.847,$$

the per-step expansion factor. Cubing (to account for the three indices of the spine) gives the spine growth rate

$$\alpha_3^3 = \frac{1}{(\sqrt[3]{2} - 1)^3} \approx 56.948.$$

The consecutive-ratio table shows ratios approaching 56.95 from below:

n	$D_3(n, n, n)$	ratio to previous
1	13	—
2	409	31.46
3	16081	39.32
4	699121	43.48
5	32193253	46.05
6	1538743249	47.80
7	75494983297	49.06
8	3776339263873	50.02

5 The Defining Cubic

The number α_3 satisfies a clean algebraic equation. Since $t_* = \sqrt[3]{2} - 1$ is the dominant positive pole, the reciprocal-pole definition gives

$$\alpha_3 = \frac{1}{t_*}.$$

Hence

$$\frac{1}{\alpha_3} = \sqrt[3]{2} - 1,$$

so

$$1 + \frac{1}{\alpha_3} = \sqrt[3]{2}.$$

Multiplying through by α_3 gives

$$\alpha_3 + 1 = \sqrt[3]{2} \alpha_3.$$

Cubing both sides,

$$(\alpha_3 + 1)^3 = 2\alpha_3^3.$$

Expanding the left-hand side gives

$$\alpha_3^3 + 3\alpha_3^2 + 3\alpha_3 + 1 = 2\alpha_3^3,$$

and therefore

$$\boxed{\alpha_3^3 = 3\alpha_3^2 + 3\alpha_3 + 1}.$$

This is the cubic analogue of the silver mean's defining quadratic $\alpha_2^2 = 2\alpha_2 + 1$. The coefficients on the right-hand side are the *binomial coefficients* $\binom{3}{1}, \binom{3}{2}, \binom{3}{3}$, the same coefficients that appeared in the denominator of the generating function and that count the king's step types by dimensionality. This is the structural result we have been chasing. The defining equation of the per-step expansion factor in d dimensions is

$$\alpha_d^d = \binom{d}{1}\alpha_d^{d-1} + \binom{d}{2}\alpha_d^{d-2} + \cdots + \binom{d}{d-1}\alpha_d + \binom{d}{d}. \quad (3)$$

Equivalently, $(\alpha_d + 1)^d = 2\alpha_d^d$, and so $\alpha_d = 1/(\sqrt[d]{2} - 1)$. The first few cases:

d	Defining equation	α_d (closed form)	α_d (numerical)
1	$\alpha_1 = 1$	1	1.0000
2	$\alpha_2^2 = 2\alpha_2 + 1$	$1 + \sqrt{2}$	2.4142
3	$\alpha_3^3 = 3\alpha_3^2 + 3\alpha_3 + 1$	$1 + \sqrt[3]{2} + \sqrt[3]{4}$	3.8473
4	$\alpha_4^4 = 4\alpha_4^3 + 6\alpha_4^2 + 4\alpha_4 + 1$	$1/(\sqrt[4]{2} - 1)$	5.2852

The spine growth rates α_d^d are 1, 5.83, 56.95, 780.28, ... growing super-exponentially in d because each higher dimension adds exponentially many more step types.

The binomial coefficients appearing as the weights of the characteristic polynomial are our reward. There are $\binom{d}{k}$ ways to choose which k of the d axes a step advances, and these counts populate the denominator of the diagonal generating function exactly.

6 Worpitzky Returns: Path-Length Decomposition

The cleanest place where Worpitzky's organising principle meets the cubical king is in the decomposition of $D_3(n, n, n)$ by path length.

A path from $(0, 0, 0)$ to (n, n, n) has a length m given by the number of moves it uses. The shortest possible path uses only triple-axis steps, giving $m = n$; the longest uses only single-axis steps, giving $m = 3n$.

Let

$$f_n(m) = \#\{\text{paths of length } m \text{ from } (0, 0, 0) \text{ to } (n, n, n)\}.$$

Then

$$D_3(n, n, n) = \sum_{m=n}^{3n} f_n(m),$$

simply because every admissible path has some definite length between n and $3n$.

The interesting question is therefore not whether such a decomposition exists, but how the individual quantities $f_n(m)$ may be computed.

Proposition 1. *Let*

$$S(x, y, z) = x + y + z + xy + xz + yz + xyz$$

be the generating polynomial of the seven legal step types of the cubical king. Then

$$f_n(m) = [x^n y^n z^n] S(x, y, z)^m,$$

the coefficient of $x^n y^n z^n$ in the m -th power of S .

Sketch. Each path of length m is an ordered sequence of m legal steps. Each legal step contributes one monomial from

$$S(x, y, z) = x + y + z + xy + xz + yz + xyz.$$

Multiplying m such monomials records the total coordinate advance of the path. The exponent of x records total motion in the x -direction, and similarly for y and z .

Thus $S(x, y, z)^m$ enumerates all length- m paths according to their total displacement, and the coefficient of $x^n y^n z^n$ selects exactly those paths terminating at (n, n, n) . $\square$

This is a fixed-length generating function: the exponent m records the number of moves, while the variables x, y, z record total coordinate advance.

For small n the path-length distributions are:

n	$m = n$	$m = n + 1$	$m = n + 2$	$m = n + 3$	$m = n + 4$	$m = n + 5$	$m = n + 6$	total
0	1							1
1	1	6	6					13
2	1	24	114	180	90			409
3	1	60	690	2940	5670	5040	1680	16081

Several structural features appear immediately.

Shortest paths ($m = n$). The single value 1 in the first column counts the unique path using only $(1, 1, 1)$ steps.

Longest paths ($m = 3n$). The final column counts paths using only single-axis moves. Such a path contains exactly n steps in each coordinate direction, so the number of interleavings is the trinomial coefficient

$$\binom{3n}{n, n, n} = \frac{(3n)!}{(n!)^3},$$

giving the sequence (OEIS A006480):

$$1, 6, 90, 1680, \dots$$

Asymmetry. The rows are not symmetric because the seven move types have different efficiencies with respect to path length: the step $(1, 1, 1)$ advances three coordinates at once, whereas the single-axis moves advance only one.

The more rewarding structure lies in the interior diagonals.

A non-trivial interior spine. The interior diagonal

$$1, 6, 114, 2940, 87570, \dots$$

is not accidental. It is the sequence OEIS A274786, [7]¹ listed as the diagonal of the rational function

$$\frac{1}{1 - (wxz + wy + wz + xy + xz + y + z)}.$$

¹OEIS A274786 links this sequence to work on diagonals of rational functions, including Bostan, Boukraa, Maillard, and Weil.

Equivalently,

$$a(n) = \sum_{j=0}^{2n} (-1)^j \binom{2n}{j} \binom{j}{n}^3.$$

This confirms that the path-length table has structure in its interior, not only at its trivial boundary columns. The shortest-path column is geodesic, the longest-path column is multinomial, but the internal spines belong to the broader theory of diagonals of rational functions, [8]. That is precisely the point at which the cubical king stops being merely a three-dimensional extension of the chessboard problem and begins to touch the arithmetic of rational-function diagonals.

7 Summing up

The promotion from two-dimensional to three-dimensional king-walks makes several previously hidden structures visible.

First, the silver-mean structure of the 2D problem is revealed to be only one rung of a larger algebraic ladder. The 3D analogue

$$\alpha_3 = 1 + \sqrt[3]{2} + \sqrt[3]{4}$$

satisfies a cubic, the 4D analogue satisfies a quartic, and so on. These are not metallic means or quadratic units; rather they share a common defining structure:

$$(\alpha_d + 1)^d = 2\alpha_d^d.$$

The dimensionality of the lattice promotes the algebraic degree of the growth constant.

Second, the appearance of binomial coefficients in the characteristic equation reveals why Worpitzky's triangle belongs naturally beside the cubical king. Worpitzky's triangle organises power sums through binomially weighted inheritance. The cubical king instead organises paths by dimensionality of motion: single-axis, double-axis, and triple-axis steps. The defining cubic

$$\alpha_3^3 = 3\alpha_3^2 + 3\alpha_3 + 1$$

contains coefficients

$$\binom{3}{1}, \binom{3}{2}, \binom{3}{3},$$

which count exactly the number of legal step types of each dimensionality: three single-axis moves, three double-axis moves, and one triple-axis move. Both structures arise from the same combinatorial source: the binomial expansion of

$$(1 + x)^d,$$

which counts subsets of a d -element set by size. In this sense, Worpitzky's identity turns powers into binomials, whereas the cubical king's defining equation turns the spine growth rate into binomially weighted powers of itself.

Third, the path-length decomposition

$$D_3(n, n, n) = \sum_{m=n}^{3n} f_n(m)$$

shows that the cubical king possesses nontrivial internal structure beyond its shortest and longest paths. The interior diagonals already produce known sequences associated with diagonals of rational functions, suggesting that the path-length table contains arithmetic structure well beyond straightforward multinomial counting.

8 Closing Thought

The two-dimensional chessboard gave us 204 squares of all sizes via Worpitzky's row 2, and 48,639 corner-to-corner king-walks via the two-dimensional central Delannoy number. We have now in three dimensions: the $8 \times 8 \times 8$ cubical block contains 1296 small cubes of all sizes (via Worpitzky's row 3), and admits $D_3(7, 7, 7) = 75,494,983,297$ corner-to-corner king-walks. Both counts are generated by the same combinatorial machinery: subsets of d items by size, with binomial-coefficient weights.

The three-dimensional king is the natural continuation of the chessboard problem where the algebra escalates from quadratic to cubic, the geometry from D_4 square symmetry to D_8 cubic symmetry, and the combinatorics from three step types to seven. Worpitzky's organisation of power sums by binomial weights is precisely the right scaffolding for this escalation, and the binomial coefficients that appear in the cubic's characteristic polynomial are its fingerprint.

Exercise. Compute $D_3(2, 2, 2) = 409$ directly by summing the seven predecessor values, given $D_3(2, 2, 1) = D_3(2, 1, 2) = D_3(1, 2, 2)$ etc. You will need to compute the smaller D_3 values first.

Exercise. Verify the path-length distribution for $n = 2$ by direct enumeration. There should be exactly one path of length 2, twenty-four paths of length 3, one hundred and fourteen paths of length 4, one hundred and eighty paths of length 5, and ninety paths of length 6. The length-2 case is easy (the unique double-triple). The length-6 case is the central trinomial coefficient $\binom{6}{2, 2, 2} = 90$.

Exercise. Set up the four-dimensional king. He has $2^4 - 1 = 15$ legal forward step types. Show that the central spine $D_4(n, n, n, n)$ satisfies a recurrence with 15 terms. Compute $D_4(1, 1, 1, 1)$ and $D_4(2, 2, 2, 2)$ by direct enumeration. Verify the expected growth rate $\alpha_4 = 1/(\sqrt[4]{2} - 1)$ raised to the fourth power gives roughly 780.

Exercise. Show that the asymptotic expression

$$D_d(n, n, \dots, n) \sim \frac{\alpha_d^{dn}}{C_d n^{(d-1)/2}}$$

holds for some constant C_d , where $\alpha_d = 1/(\sqrt[d]{2} - 1)$. Hint: the saddle-point method applied to the diagonal extraction of the multivariate generating function.

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